A Natural-Language Interface for Healthcare Data



HDA

Presented by

Thomas Alfio Drury

Supervisor: Prof. Giovanni Morana

Agenda

What HDA is, why a hospital needed it, and what it proved.

01

Context

The form behind every discharge, and why one simple question takes a month.

02

Solution

Ask in plain language, narrow the answer, and let the app offer more.

03

Conclusion

How it was tested, what it delivers, and what is still missing.

From the form behind every discharge to an answer you can trust.

One record, every discharge

A single form, filled out at every hospital discharge

One form per stay

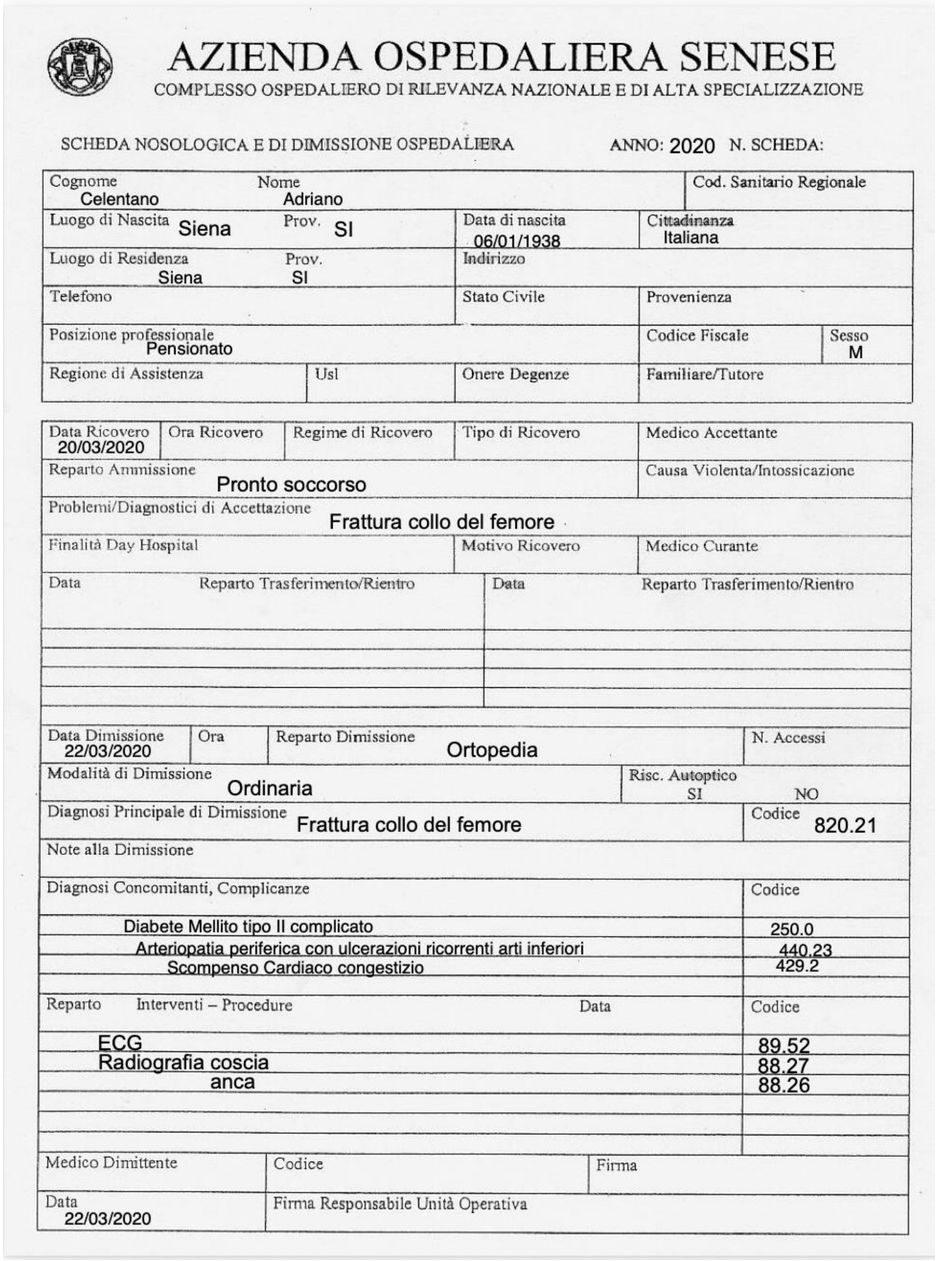
The Scheda di Dimissione Ospedaliera, with the same legal validity as the medical record.

DRG and ICD-9-CM

The diagnosis-related group, plus coded diagnoses and procedures.

Demographics

Age, sex, residence, and the dates of every admission and discharge.



AZIENDA OSPEDALIERA SENESE

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SCHEDA NOSOLOGICA E DI DIMISSIONE OSPEDALIERA — ANNO: 2020 N. SCHEDA:

Cognome: Celentano	Nome: Adriano			Cod. Sanitario Regionale
Luogo di Nascita: Siena	Prov.: SI	Data di nascita: 06/01/1938	Cittadinanza: Italiana	
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Telefono		Stato Civile	Provenienza	
Posizione professionale: Pensionato			Codice Fiscale	Sesso: M
Regione di Assistenza	Usl	Onere Degenze	Familiare/Tutore	

Data Ricovero: 20/03/2020	Ora Ricovero	Regime di Ricovero	Tipo di Ricovero	Medico Accettante
Reparto Ammissione: Pronto soccorso				Causa Violenta/Intossicazione
Problemi/Diagnostici di Accettazione: Frattura collo del femore				
Finalità Day Hospital			Motivo Ricovero	Medico Curante

Data	Reparto Trasferimento/Rientro	Data	Reparto Trasferimento/Rientro

Data Dimissione: 22/03/2020	Ora	Reparto Dimissione: Ortopedia	N. Accessi
Modalità di Dimissione: Ordinaria			Risc. Autoptico: SI NO
Diagnosi Principale di Dimissione: Frattura collo del femore			Codice: 820.21
Note alla Dimissione			

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Reparto	Interventi – Procedure	Data	Codice
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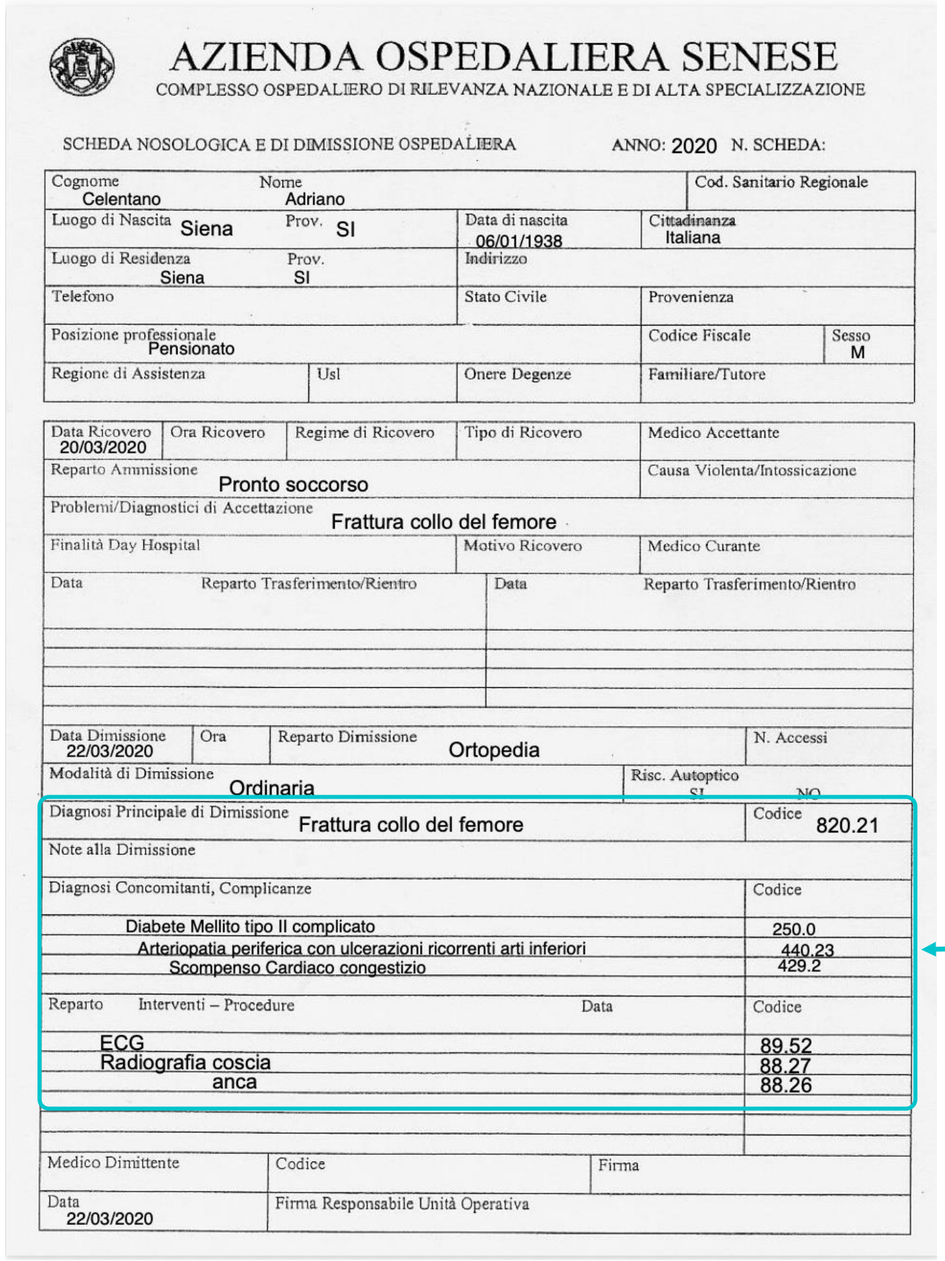
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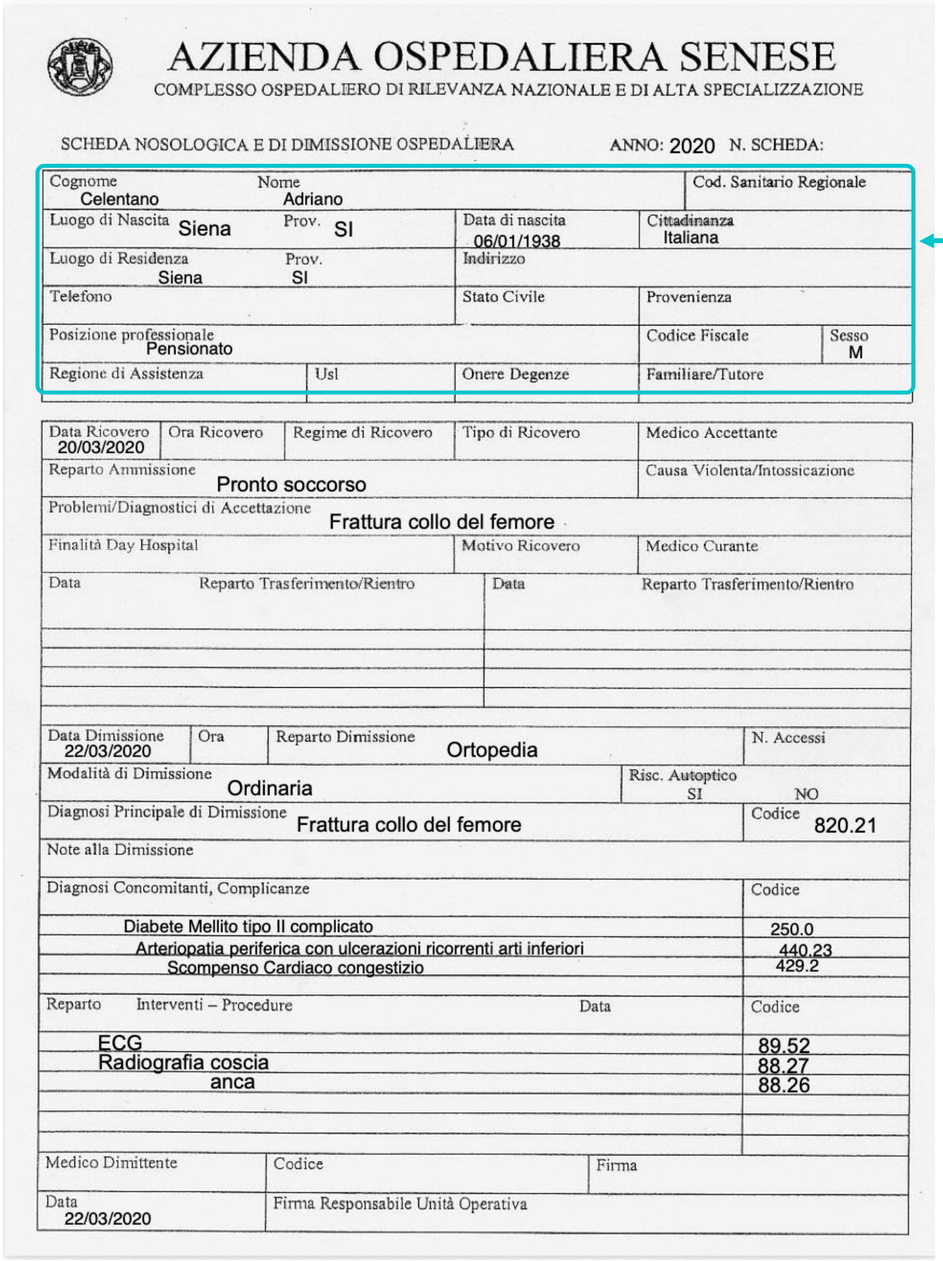
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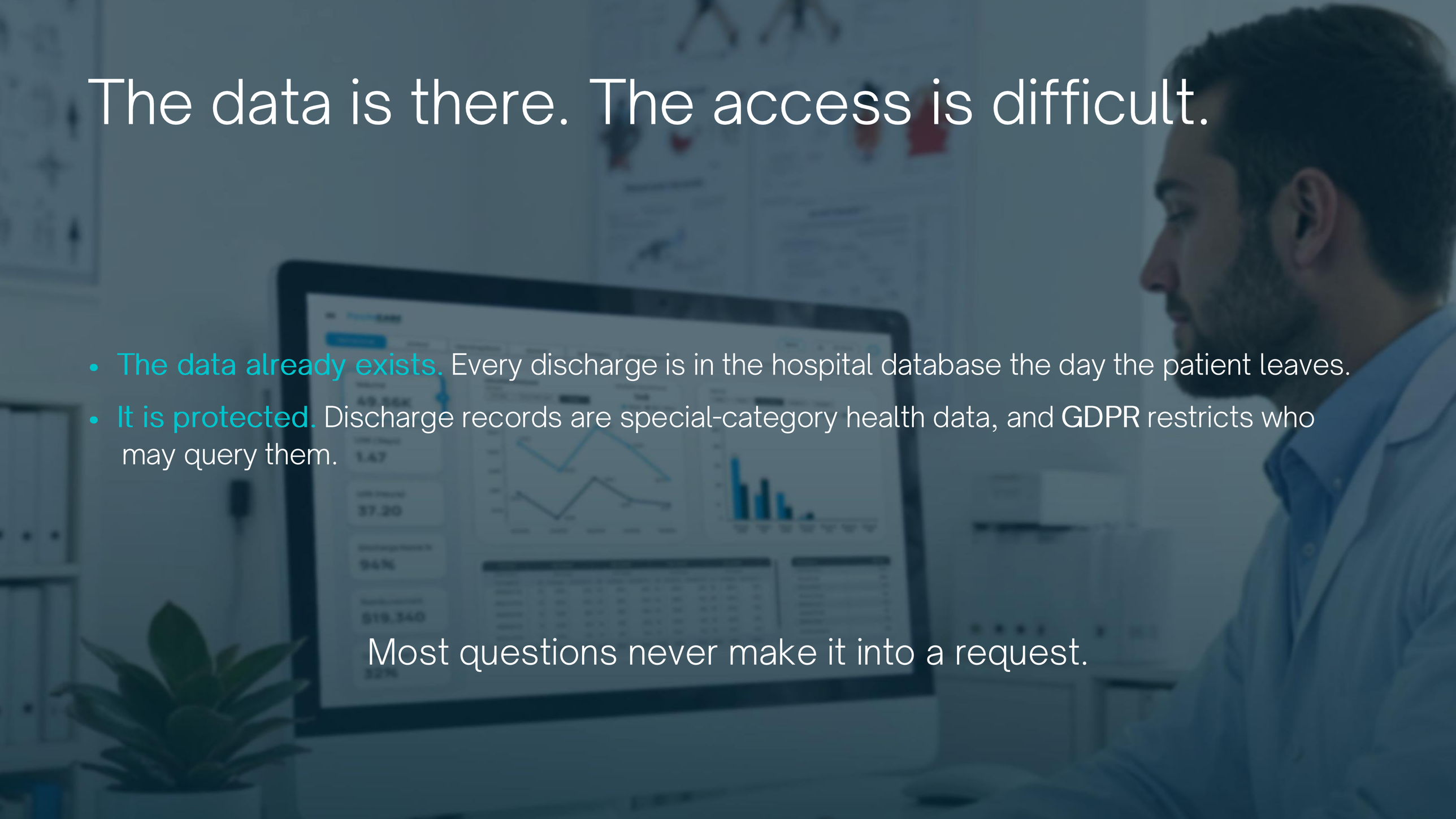
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The data is there. The access is difficult.



- The data already exists. Every discharge is in the hospital database the day the patient leaves.
- It is protected. Discharge records are special-category health data, and GDPR restricts who may query them.

Most questions never make it into a request.

Before and after

The same question, asked two ways.

Now

about a month

- Ask Controllo di Gestione for the results.
- Wait for the release in the following month.
- Read it in the structure they chose.
- A new question starts the cycle again.

With HDA

about ten seconds

- Ask the database yourself, in plain Italian.
- Read the answer in about ten seconds.
- Filter it and chart it as the question develops.
- A new question is simply the next question.

The monthly reporting stays where it is. HDA covers the questions in between.

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Possible solutions

Why the obvious AI fix isn't safe enough

A confidently wrong answer
is worse than no answer at all.

Text-to-SQL

The model writes the SQL directly.

It runs untouched - no verifier to catch a bad query.

The best published systems still get it wrong 1 time in 4.

What it must do instead

Checkable - show what it understood, before it answers.

Honest - unsure? Refuse, never guess.

Local - no patient data ever leaves the hospital.



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The idea

The model only interprets. It picks an action that already exists and fills its parameters; it never writes a query.

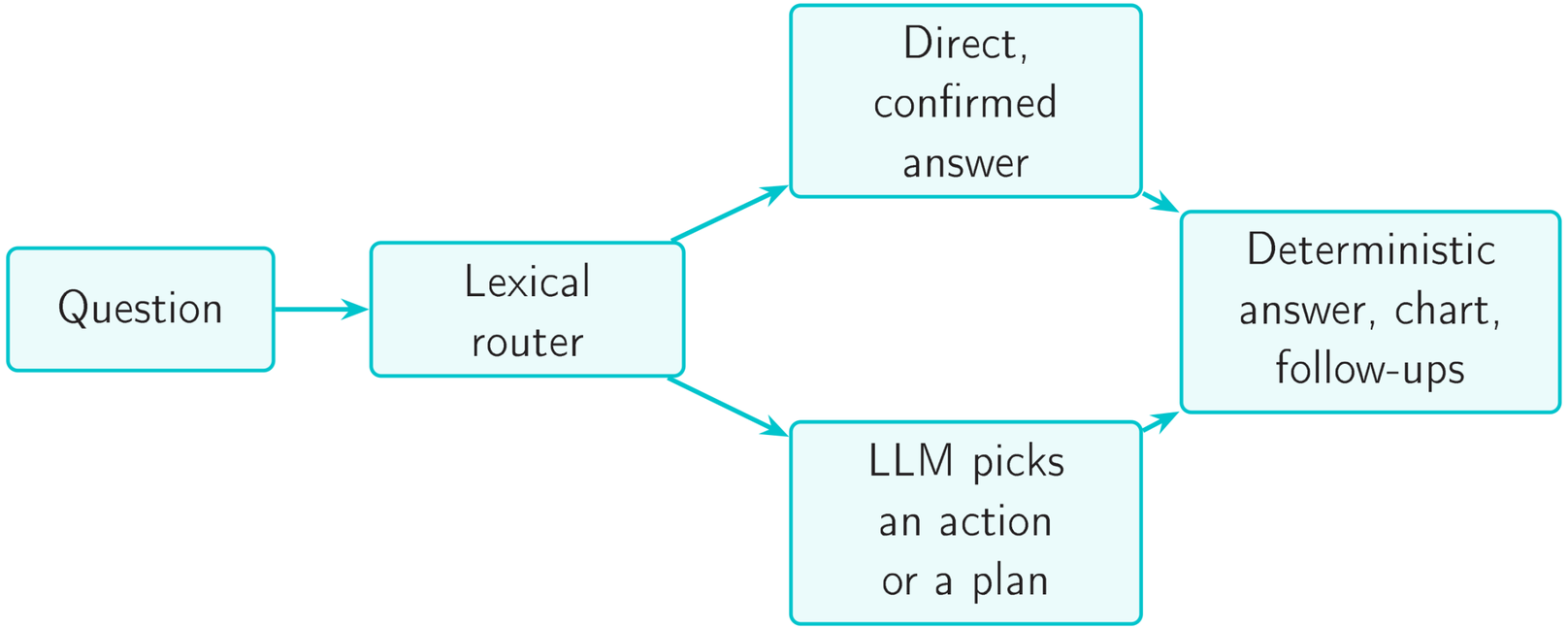
Nothing leaves the building - the model runs on the hospital's own hardware. Small patient groups are suppressed automatically, and an unclear question is refused, never guessed.



The LLM, kept on a short leash

Called only when the lexical router cannot match the question, and only to choose a tested action and fill them with defined parameters

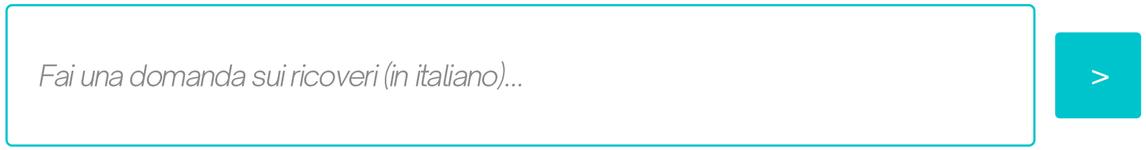
Hosted inside the hospital on a local Qwen 9.5B.



In practice

01 Ask, get an answer

Type a question in plain language - the app answers directly, no query language needed.



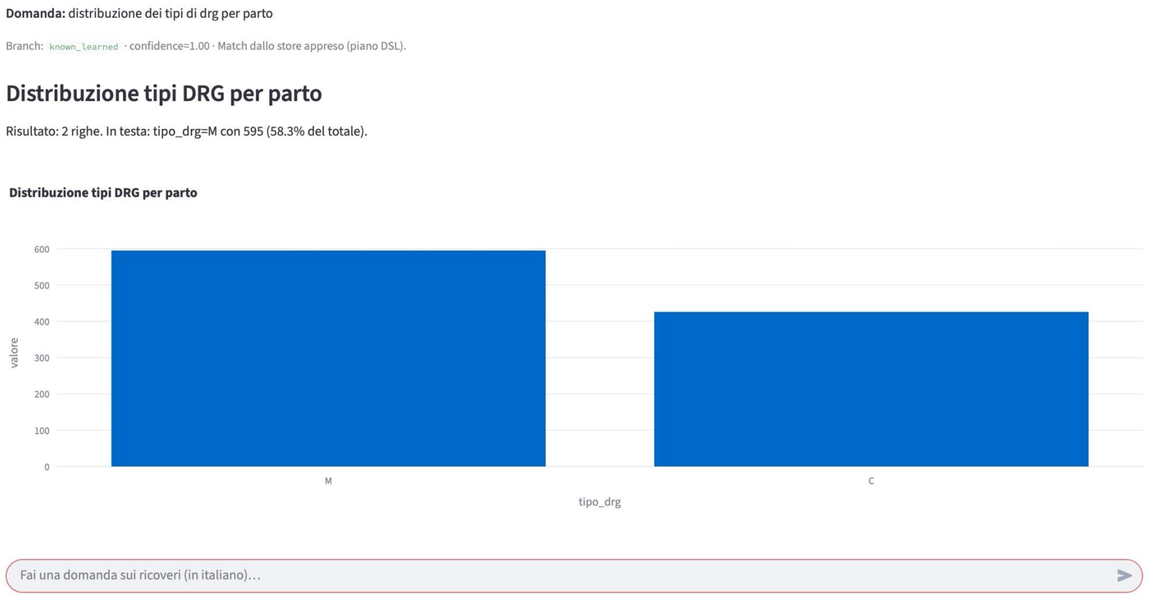
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Filter by sex, age band, year or DRG, and the answer narrows without asking again.

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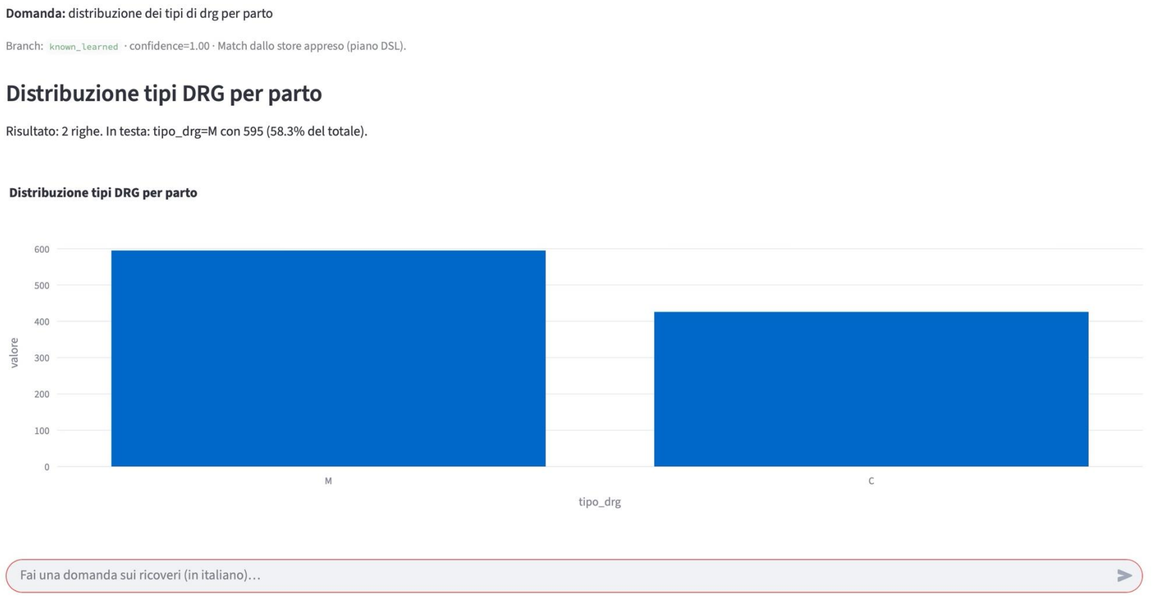
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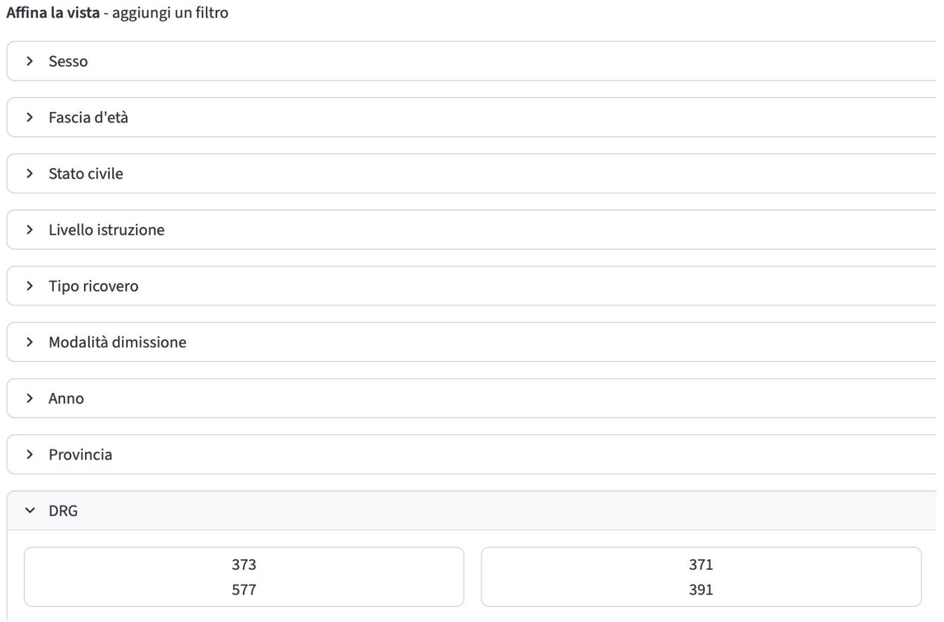
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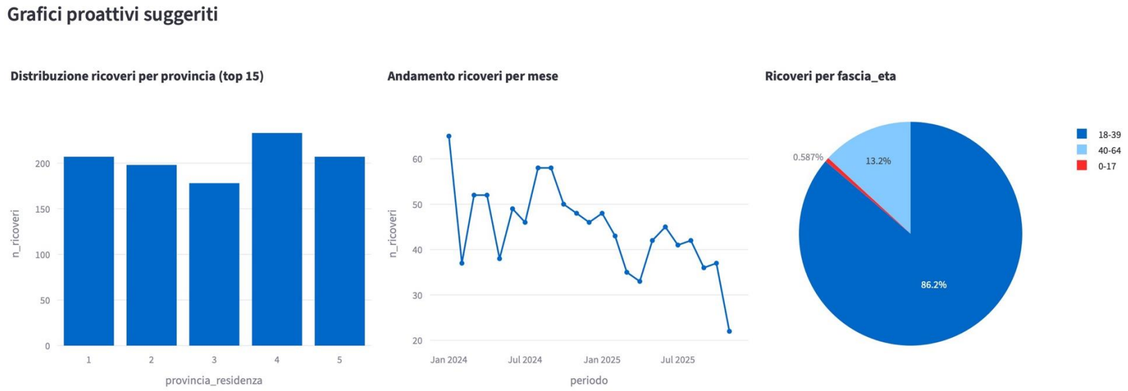




See more

03 Charts before you ask

Up to 3 further charts on the same data, offered rather than requested.



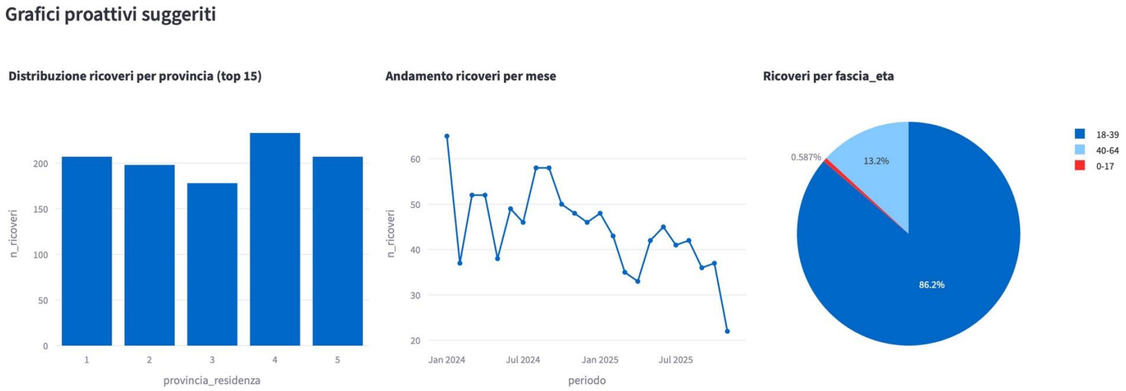
04 Follow-up questions

Three questions it suggests next, so the thread carries on.

See more

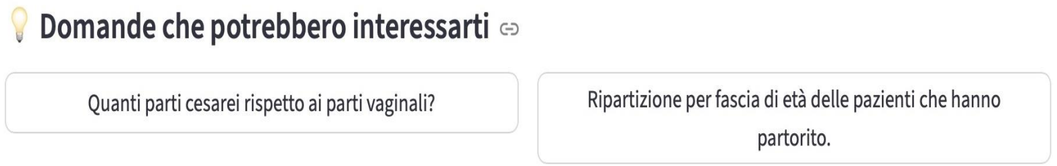
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How it was tested

25 questions against the same 18,000-discharge extract



Real data

An 18,000-discharge extract - not a synthetic benchmark.



5 question families

Five families of five questions each, from direct actions to questions the system is expected to refuse.



Wrong branch = failure

Reaching the wrong branch counts as a failure, even when the final number is right.



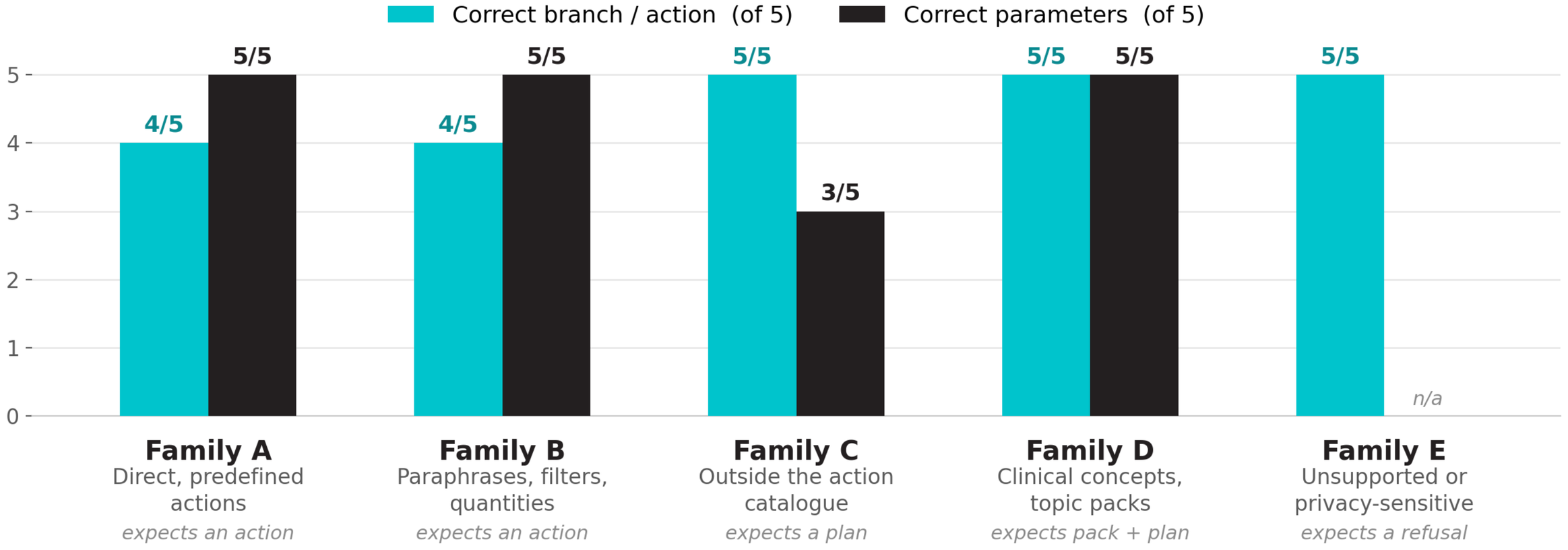
Correct parameters

The right branch with the wrong filter still fails: every value the model fills in is scored on its own.

Results: 21 of 25

What each family tests, and how it scored out of 5





Limits

Odd phrasing can point it at the wrong analysis, and a confident match can still be wrong. Privacy is checked per answer, not per session.

...and what's next

A clinician's words and the ICD-9-CM/DRG codes remain unlinked, today. The fix: a clinical category layer in between - shown here for a hip-fracture example.

Each layer narrows the clinician's words towards the codes the database actually stores. The middle three are what HDA is missing today.

Until that layer exists, HDA answers what it can verify - and refuses the rest.

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The real difference

A ward manager used to wait a month for a simple answer. Now they type a question in plain language and get one they can trust, in seconds.

That's the real solution: giving time back to the people who ask the questions.

Thank you for your attention!



Thomas Alfio Drury - Supervisor: Prof. Giovanni Morana,
University of Catania